

YEOW Jun Wei

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EDUCATION

Doctor of Philosophy

Aug'23 – May'27 (expected)

Nanyang Technological University (NTU), Singapore

Thesis Title: “Advancing Environmental Acoustic Intelligence: Robust, Geometric, and Lightweight Frameworks for 3-D Sound Event Localization and Detection”

Awards: Recipient of NTU Research Scholarship

B. Eng. (Hons) in Electrical and Electronic Engineering

Aug'19 – Jul'23

Nanyang Technological University (NTU), Singapore

CGPA: 4.50/5.00 (Graduated with Honours, Highest Distinction)

Second Major: Business (Specializing in Investments)

Relevant Coursework: Digital Signal Processing, Machine Learning, Deep Learning

RESEARCH EXPERIENCE

Label-Efficient Sound Event Localization and Detection

Jan'26 – Present (PhD Student)

Digital Signal Processing Laboratory, NTU

Supervisor: Prof. Gan Woon Seng

- Investigating novel methodologies to significantly reduce the reliance on strongly annotated data, advancing label-efficient training paradigms for Sound Event Localization and Detection (SELD) models
- Designing robust frameworks tailored to mitigate the extreme class imbalance and long-tailed distributions inherently found in real-world environmental acoustic datasets

Deep Ear for Hearables

Aug'23 – Dec'25 (PhD Student)

Digital Signal Processing Laboratory, NTU

Supervisor: Prof. Gan Woon Seng

- Assisted research and development of lightweight, real-time SELD frameworks for deployment on resource-constrained edge devices and hearable technology
- Demonstrated a live, polyphonic SELD system to an international audience at the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2024 in Korea

TEACHING EXPERIENCE

Nanyang Technological University

Undergraduate Research Supervisor, EE4080 Final Year Project

Aug'23 – Present

Supervised original research projects with real-world applications for final-year undergraduate students, guiding them through the complete research lifecycle from conceptualization to execution

Teaching Assistant, IE0005 Introduction to Data Science and Artificial Intelligence

Jan'26 – May'26

Led comprehensive instructional lessons and hands-on lab sessions, effectively breaking down foundational data science and artificial intelligence concepts for undergraduate students

Teaching Assistant, IE4727 Web Application Design

Aug'25 – Dec'25

Facilitated design electives, mentoring students in building, designing, and deploying modern web applications

SELECTED PUBLICATIONS

- J.W. Yeow, E.L. Tan, S. Peksi, W.S. Gan, “MAGENTA: Magnitude and Geometry-Enhanced Training Approach for Long-Tailed Sound Event Localization and Detection”, accepted in IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2026

2. J.W. Yeow, E.L. Tan, S. Peksi, W.S. Gan, "WINTER: Wrapped Interval Normalization for Elevation Representation in Stereo 3-D Sound Event Localization and Detection", IEEE Signal Processing Letters, 2026
3. J.W. Yeow, E.L. Tan, S. Peksi, W.S. Gan, "Environmental Acoustic Intelligence through Sound Event Localization and Detection: A Review", npj Acoustics, 2025
4. J.W. Yeow, E.L. Tan, S. Peksi, W.S. Gan, Q. Huang, "Towards Robust Stereo 3-D SELD: A Study of Perceptual Features and Data Augmentation", in Asia Pacific Signal and Information Processing Association Annual Conference, 2025: Singapore
5. J.W. Yeow, E.L. Tan, S. Peksi, Z.T. Ong, W.S. Gan, "Enhancing Situational Awareness in Wearable Audio Devices Using a Lightweight Sound Event Localization and Detection System", in 2nd Audio Engineering Society International Conference on Headphone Technology, 2025: Finland
6. J.W. Yeow, E.L. Tan, J. Bai, S. Peksi, W.S. Gan, "Enhancing 3-D Sound Event Localization and Detection with Distance Estimation using Reverberation and Spatial Coherence Features", IEEE Sensors Journal, 2025

WORK EXPERIENCE

Silicon Labs

May'22 – Dec'22

Product Test Engineer Intern

- Collaborated closely with cross-functional engineering teams to develop and refine comprehensive test scripts
- Executed rigorous product testing to validate components, ensuring seamless integration for new product lines

Siemens Mobility

Jun'21 – Dec'21

Data Engineer Intern

- Used a technology stack of Hive, NiFi, Pyspark, Jenkins, Shell, and Python Scripts to create and maintain scalable and reliable data pipelines
- Responsible for improving existing Python and Shell scripts, reducing the time and resources needed by up to 50%

CO-CURRICULAR ACHIEVEMENTS

NTU Hall 11

Honorary General Secretary

Aug'20 – May'21

- Chief administrative officer of the hall management committee, responsible for internal operations, documentation, and communications

SKILLS

- **Languages:** English (native), Mandarin Chinese (fluent)
- **Programming Experience:** MATLAB, C, Python, SQL, Shell
- **Technical Skills:** Machine Learning, Deep Learning, Research, Digital Signal Processing, Artificial Intelligence